

Question 1: Ethics, Law and professional responsibility.

a) Difference between Ethics, Law and professional responsibility.

Ethics refers to moral principles that help people decide what is right or wrong. In computing, ethics guides developers to protect privacy, security and user interest.

Law refers to rules created and enforced by the government. Breaking law can result in legal punishment.

professional responsibility refers to the duties and responsibility that a computing professional has toward users, employers, colleagues and society.

(c) Major principles of software Engineering Ethics:

Software engineering ethics provides guidelines for responsible professional behaviour -

1. public safety: software should not unnecessarily put users at risk.
2. privacy: Developers should protect users personal and sensitive information.
3. security: software should be designed to prevent unauthorized access.
4. Honesty: Engineer should honestly communicate about software & limitation and risks.

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Question 2: Ethical Frameworks and a software privacy dilemma.

(a) Utilitarianism vs Deontology

Utilitarianism judges an action mainly by its consequence. An action is considered ethical if it ~~pro~~ produces the greatest overall benefit for the greatest number of people.

Deontology, especially Kantian ethics, focuses on duties, rules and rights rather than only consequence. From this perspective, users have a right to privacy and should not be treated merely as a means to increase company profit.

(b) Virtue Ethics:

Virtue Ethics focuses on the character and moral qualities of a person rather than only rules or consequences.

A software engineer should demonstrate virtues such as -

(i) Honesty: clearly explain how user data is being used.

(ii) Fairness: treat users equally.

(iii) Responsibility: consider the consequences of data collection.

(iv) Courage: raise concern when a company's decision may harm users.

Question-3: Explainability and Black-Box AI.

(a) Importance of transparency and Explainability.

Transparency means providing information about how an AI system works, what data it uses, and how decisions are made.

These are especially important in high-stakes areas such as

- (i) credit lending
- (ii) Healthcare
- (iii) Employment
- (iv) Criminal justice

Transparency and explainability help

to

- (i) Detect unfair or biased decision
- (ii) Increase accountability
- (iii) Allow people to challenge incorrect decision

(A) Build public trust

(B) Societal Harms of Black-Box AI

A black-box AI system is difficult to understand or explain. It can (i) create several harm. (ii) first discrimination. If the system contains biased training data, it may unfairly reject or disadvantage certain groups.

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Second, lack of accountability: If nobody can explain why the AI made a decision, it becomes difficult to determine who is responsible when the decision causes harm.

Question - 4: ACM code of Ethics and
prof. professional decision making.

(a) Key principle of the ACM code of Ethics.

The ACM code of Ethics provides principles that guide computing professionals toward responsible behaviour.

Important principles include -

(i) contribute to society and human well-

- (i) Avoid harm
- (ii) Be honest and trustworthy
- (iv) Be fair and take action not to discriminate.
- (v) ~~Respect~~ Respect privacy
- (vi) Honor confidentiality
- (vii) maintain professional competence.

(b) Relevant ACM principle in the scenario -

The ~~eng~~ engineer should consider several principle.

Avoid harm: Releasing a system with serious security weakness may harm ~~eng~~ users.

Respect privacy: Security weakness may expose users personal information.

Design and implement secure system:

Security problems discovered during testing should be addressed before release when they create serious risks.

Question - 5: Whistleblowing and professional

Dissent

(a) professional Dissent and whistleblowing:

professional dissent means expressing disagreement with a professional

decision or practice because the person

believes it is unsafe unethical or wrong

For example, a cybersecurity analyst may disagree with management's decision

to ignore a serious vulnerability.

(c) Risks to the whistleblower:

Whistleblower may face several risks -

(i) Professional risk: Loss of promotion, demotion or termination.

(ii) Legal risk: possible legal ~~risky~~ disputes depending on how information is disclosed.

(iii) Organizational risk: conflict with management or ~~colle~~ colleague.

(iv) Personal risk: stress, anxiety, financial difficulties, or damage of reputation.

Question - 6 : privacy versus social
Benefit.

(a) Society and human well-being vs
respect for privacy

The ACM principle "contribute to
society and human well-being" encourage
computing professionals to use technology
for social benefit.

In the scenario, analyzing mobile-phone
location data may help authorities identify
disease information.

These principles may conflict because
using location data can provide public-
health benefits but may also reveal
where people live, work, study and
travel -

(b) Ethical principles for personal data:

Four important principles include.

(i) Informed consent: people should understand how their data will be used.

(ii) Purpose limitation: data should be collected for a specific legitimate purpose.

(iii) Data minimization: Only necessary data should be collected.

(iv) Security: Data should be protected against unauthorized access.

Question - 7: copyright, patents, and Trademark.

(a) Difference between copyright, patents and trademark.

Copyright protect original creative work such as source code, documentation, graphics, and other original expressions.

Patents protects eligible new invention or ~~technie~~ technical innovation, subject to applicable law and requirements.

Trademark protects names, logos, symbols and other identifiers that distinguish a product or organization.

(b) Appropriate protection.

The startup should use different forms of intellectual property protection for different assets — assets.

Source code: copyright is generally appropriate because it protects the original expression in the code.

novel technical invention: A patent may be appropriate if the invention satisfies applicable patent requirements.

Question-8: Open-source and proprietary software Licensing.

(a) Open-source vs proprietary software

Open-source software is software whose source code is made available under a license that permits certain form of use, modification and redistribution

<u>Open-source</u>	<u>proprietary</u>
(i) source code is available under the license	(i) source code is generally restricted
(ii) can encourage collaboration	(ii) Greater commercial control
(iii) Researchers can inspect the code	(iii) Users have limitation of modification rights

© Importance of understanding License obligations

Developers must understand software licenses before using third-party code.

Different licenses may have different requirements regarding —

i) Attribution

ii) Modification

iii) Redistribution

iv) Source-code availability

v) Commercial use.

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Question-9: Responsible Disclosure.

(a) Responsible disclosure -

Responsible disclosure is the ethical process of reporting a security vulnerability to the responsible organization so that it can investigate and fix the problem before or while coordinating public disclosure.

Responsible disclosure is different from unauthorized exploitation.

A security researcher should demonstrate the vulnerability safely without necessarily accessing or damaging real users' data.

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(b) Ethical guideline for the
Researcher.

The researcher should -

- i) confirm the vulnerability safely.
- ii) Avoid necessary access to personal information.
- iii) Avoid damaging or disrupting the system.
- iv) Identify the responsible organization.
- v) Report the vulnerability securely.
- vi) provide enough technical information to reproduce the problem.
- vii) Give the organization reasonable time to fix it.

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Question - 10 : Ethical Hacking vs

cybercrime.

(a) Difference between Ethical Hacking and cybercrime.

Ethical hacking is authorized security testing performed to identify vulnerabilities and improve security.

cybercrime involves unauthorized or illegal activities using computers or networks, such as stealing information, damaging system, or accessing accounts without permission. The main difference is authorized and purpose.

⑥ condition of Ethical Hacking

Hacking may be considered ethical when it satisfies important condition.

i) Authorization: proper permission

must exist.

ii) Scope: The tester must only test approved systems.

iii) Purpose: The purpose should be improving security.

iv) Proportionality: Testing should not cause unnecessary damage.

v) Privacy: personal information must be protected.

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Question- 11: Algorithm bias in Recruitment.

⊙ Algorithm Bias and its causes.

Algorithm bias occurs when an algorithm produce systematically unfair or discriminatory results for certain individual or groups.

possible cause include -

- i) Biased training data: Historical data may contain discrimination.
- ii) Historical discrimination: previous hiring decision may already unfair.
- iii) proxy variables: variable such as location may indirectly represent demographic characteristics.
- iv) sampling problem: The training dataset may not properly represents all group.

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(c) measures for improving fairness -

Technical measures include

- i) Testing datasets for bias
- ii) using representative training data
- iii) measuring outcome across different groups.
- iv) Regularly auditing the model
- v) Monitoring the system after deployment

Organization measures include -

- i) Human review of important decision
- ii) clear fairness policy
- iii) Independent audit
- iv) employee training
- v) Mechanism for applicants to challenge decisions.

question - 12.

Accountability in Autonomous system.

Q) Accountability in Autonomous Systems

Accountability means that people and organizations involved in developing, deploying, or operating an autonomous system should be responsible for their decision and actions.

Responsibilities may exist at different stages, including

- i) Requirements
- ii) Data collection
- iii) Model design
- iv) Software development
- v) Testing
- vi) Deployment
- vii) Monitoring
- viii) Incident response

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(c) Why "The AI made the Decision"
Is Not Enough -

Saying "the AI made the decision"
does not automatically remove human
responsibility.

Human Selected -

- i) System requirement
- ii) Training data
- iii) Model
- iv) Software
- v) Testing method
- vi) Deployment condition

If Developers fail to test a system
properly or management knowingly
deploys an unsafe system, they may
still have responsibility for the
resulting harm.

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Question - 13:

Ethical Leadership in IT Organizations

(a) Ethical Leadership:

Ethical leadership means leading an organization in a way that promotes honesty, fairness, responsibility, transparency, safety and respect for people.

In an IT organization, ethical leaders should not focus only on speed and profit. They should also consider -

- i) security
- ii) privacy
- iii) user safety
- iv) Employee concerns
- v) Legal requirements
- vi) social impacts

⑥ Establishing an Ethical culture
leaders can create an ethical
culture through:

policy: Establish clear ethical
and Security policies.

Incentive: Reward responsible
behaviour, not only rapid releases.

Training: provide regular ethics,
privacy, and security training.

communication: Encourage
employees to communicate concerns.

Transparency: clearly explain important
decisions.

Accountability: Take action when ethical
rule are violated.

Question - 14: Informed consent?

(a) Informed consent

Informed consent means that users understand what data will be collected, why it will be collected, how it will be used and what consequences may result before agreeing to the collection.

For consent to be meaningful, users should receive clear and understandable information.

In the given scenario, users are not clearly told that precise location data will be continuously collected and shared with third parties.

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(c) Characteristic of Ethical consent

Ethical consent should have the following characteristic -

- i) Clarity: Information should be easy to understand.
- ii) Transparency: The organization should clearly explain data collection and use.
- iii) Voluntariness: Users should not be unfairly forced into consent.
- iv) Specificity: Consent should clearly identify what data is being collected and why.

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Question - 16: copyright and patents.

(a) copyright vs patents
copyright protects original creative expression

In software, this may include source code, documentation, graphics and other original works.

patents protects eligible inventions and technical innovations when the applicable legal requirements are satisfied.

The main difference is that copyright generally protects the expression of an idea, while a patent can protect an eligible invention or technical solution.

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(b) Assets suitable for protection.

The company's assets can require different type of protection.

i) The source code: primarily copyright,

ii) Geographical user interface: copyright may protect original visual expression, subject to applicable law.

iii) New image-compression technique: may be suitable for patent protection if legal requirements are met.

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Question-20: Integrated Scenario-
privacy, security and business
pressure.

① Ethical, legal, professional, privacy
and cybersecurity issues.

The startups situation involves several
important issues.

Ethical issue: Launching a product with
a known security weakness may put
users at risk.

Legal issues: Depending on applicable
law, inadequate protection of personal
information may create legal responsibili-
-ties or consequences.

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(d) Ethical responsible course of Action

The company should not immediately launch the product with a known serious vulnerability. A responsible course of action would be -

- i) identify and understand the vulnerability.
- ii) Assess the potential risk to users.
- iii) conduct security testing again.
- iv) fix the security weakness.
- v) verify that the fix works.
- vi) protect users' personal and household data.
- vii) clearly communicate important risks and limitations.